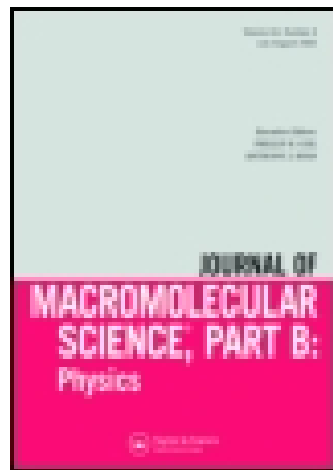




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The effect of benzoyl peroxide and divinyl benzene on the properties of crosslinked recycled polyolefin blends

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The effect of benzoyl peroxide and divinyl benzene on the properties of crosslinked recycled polyolefin blends

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Running title: The properties of BPO crosslinked PE/PP blends

Abstract:

The effect of peroxide crosslinking on the properties and morphology of recycled polyethylene (PE) / polypropylene (PP) blends was characterized. The addition of benzoyl peroxide (BPO) decreased the MFR and increased the impact strength of the recycled polymer blends. Divinyl benzene (DVB) is often used as a crosslinking agent assistant. Compared with BPO modification, the addition of BPO together with DVB improved the crosslinking efficiency and further increased the impact strength of the recycled polymer blends. The effect of BPO content on the MFR and the mechanical properties was also studied with the DVB content fixed. However, chemical crosslinking slightly reduced the thermal stability of the polymer blends. The morphology of the modified and unmodified polymer blends showed that with the addition of BPO, with or without DVB, the compatibility of the PE/PP blends was improved, resulting in enhanced impact strength.

Key words: recycled plastics; crosslinking; polyethylene/polypropylene blends; copolymer; thermal stability; morphology

Introduction

The use of plastics for packaging is increasing day by day. Every year a large amount of non-biodegradable industrial and domestic plastics are disposed of. Therefore, plastic waste has become one of the main issues of environmental concern. Recently, the recycling of plastic materials has received increasing attention because of environmental as well as economic concerns [1-3]. High-density polyethylene (HDPE) is a primary packaging material for bottle caps, along with polypropylene (PP), and is one of the most commonly recycled plastics.

Recycled plastic materials usually undergo chain scission and degradation by thermo- and photo-oxidation during the use and recycling processing [4-5], which decreases the molecular weight of the polymers as well as their mechanical properties, resulting in limited applications. Peroxide has been used to cause chain extension, branching or even crosslinking of recycled polyethylene, with higher molecular weight and enhanced properties [6]. However, recycled HDPE materials often contain a minor amount of PP. When processed with peroxide, PP degrades due to the low stability of the tertiary hydrogen atoms of macroradicals [7], which lowers the crosslinking efficiency of the PE.

Teh et al. [8] reported that one of the main problems in the modification of PE/PP blends with peroxide is the difficulty in controlling the degradation of the PP. The unwanted chain scission can be restrained by the addition of an unsaturated monomer [9]. The monomer can stabilize the polyolefin macroradical sites formed by the peroxide radicals, because the monomer converts

some of the tertiary macroradicals to the more stable secondary macroradicals and creates inter-chain block or graft copolymers by recombination [10-11]. Compared with polymeric compatibilizers, which are more expensive and usually only effective at higher concentrations, the use of the monomers showed considerable economic advantages [12].

In the research described in this paper the effect of BPO modification on the MFR and impact strength of recycled polymer blends was studied. Addition of DVB to the BPO crosslinked polymer blends further decreased the MFR and increased the impact strength of the polymer blends. The effect of BPO content on the MFR and mechanical properties was studied with fixed DVB content. Thermo-gravimetric analysis (TGA) and scanning electron microscopy (SEM) were also used to analyze the thermo-stability and morphology of the polymer blends with and without modification.

Experimental

Materials

Recycled polyethylene (containing 3-15wt% of polypropylene) coming from beverage bottle caps and minced into pieces was used. The crosslinking agent was benzoyl peroxide (BPO, supplied by Akzo-Nobel, Netherlands). Divinyl benzene (DVB, supplied by BASF, Germany) was used as a crosslinking agent assistant.

Sample preparation

Separation of PE and PP

Recycled bottle cap granules were placed in an ethanol solution ($\rho=0.92\text{g/cm}^3$) for about half an hour, with PP floating on the surface and PE subsiding to the bottom. The PP was sieved from the solution while the PE was left. The separated polymers were dried at 80°C to constant weight. Polymer blends (PE/PP=90/10wt) were obtained by precise weighing.

Extrusion

The polymer blends were premixed with all additives for about 10 min and then added to a co-rotating twin screw extruder (TSE-35A, RuiYa Extrusion System Co., Ltd., China) with $L/D=48$ and $D=35\text{ mm}$ for reactive extrusion. The extrusion temperature scale from feeding part to head was $180\text{--}210^\circ\text{C}$, while the screw speed was 300 rpm and the rate of feeding was 20 rpm. The samples were then injection molded at 80MPa, into a mold at 210°C .

Characterization

Melt flow rate (MFR)

The MFR measurement was carried out by a melt indexer (TF-2028, Tianfa Testing Machine Co., Ltd., China) with the following conditions: 190°C , 5kg. At least eight measurements of each blend were made to obtain the averages.

Mechanical properties

The tensile tests, according to ASTM D 638, were performed on a universal testing machine (model 5567, Instron Corp, USA), using specimens of 4 mm thickness and a crosshead speed of 50 mm/min. The Charpy impact strength of notched specimens was measured on an impact testing

machine (model HIT 50, Zwick GmbH & Co. KG, Germany) according to ASTM D 6110 with notched bars. All the tests were performed at room temperature and at least five measurements were made to obtain averages.

Thermogravimetric analysis (TGA)

The TGA of the injection molded samples was carried out using a TG 209 F3 (Netzsch. Co, Germany) in the temperature range between 25 and 600 °C at a heating rate of 10 °C/min in nitrogen atmosphere.

Morphology

To study the morphology, fracture surfaces obtained from impact specimens of polymer blends, with or without modification, were observed by SEM (JSM-6360LV, JEOL, Japan).

Results and discussion

The effect of BPO on the properties of polymer blends

BPO is widely used as a crosslinking agent for polyolefins. The effects of BPO on the MFR and the impact strength of the polymer blends were studied. The results are listed in Table 1. The addition of BPO lowered the MFR of the polymer blends by nearly 90%. The MFR is an indirect measurement of the changes in the average molecular weight (M_w) and the polydispersity of the sample. The decrease of MFR indicates that a crosslinked structure was formed in the system, resulting in increased molecular weight and higher melt viscosity. As for the impact strength, it is a reflection of the toughness of the material. As a result of the incompatibility of PE and PP,

leading to poor adhesion at the interface, recycled PE/PP polymer blends without modification showed low impact strength [13]. In addition, as shown by SEM (below), the molded samples consisted of foams, greatly lowered the impact strength. The addition of BPO greatly increased the impact strength of the blends. The reason is that the addition of peroxide, in addition to crosslinking the PE, according to the work of Randall et al. [14], caused the formation of PE-PP copolymer in the blends which improved the compatibility of PE and PP.

The effect of DVB on the properties of the crosslinked polymer blends

DVB is usually used as an assistant for branching or crosslinking of polymers, especially for polypropylene [15]. The effect of DVB on the MFR and impact strength of BPO crosslinked polymer blends are listed in Table 2. The MFR and the impact strength of pure HDPE were 0.33g/10min and 19.0kJ/m². The corresponding values for BPO modified HDPE/PP blends were 0.040g/10min and 14.5kJ/m². The addition of DVB further lowered the MFR but increased the impact strength of the system. As discussed above, the decrease in MFR indicates an increase in molecular weight, caused by the formation of crosslinking structure. The impact strength also can be improved by the formation of copolymer which modified the compatibility of PE and PP. In the following parts, DVB was added to the crosslinking system for the formation of higher performance material.

The effect of BPO content on the MFR and the mechanical properties of the polymer blends

Various BPO contents were used to crosslink the polymer blends with the addition of DVB content fixed at 0.5wt% of the blend. The effect of BPO together with DVB on the MFR was studied, with the results shown in Fig.1. The MFR initially decreased rapidly with the increase of BPO content, up to 0.3wt%. Further addition of BPO only slightly decreased the MFR. The decrease of MFR means an increased molecular weight being formed as a result of crosslinking. The more BPO added, the higher the degree of crosslinking formed. When the BPO content was more than 0.3%, the degree of crosslinking approached equilibrium and it increased only slightly, resulting in a small decrease of MFR.

The impact strength of the polymer blends with different BPO content, and DVB content fixed at 0.5wt%, is shown in Fig.2. The impact strength first increased with the BPO content, up to 0.3wt%. Further increase of BPO decreased the impact strength. However, the impact strength of the 0.4wt% BPO modified polymer blends was still much higher than that of the unmodified polymer blends. Crosslinking enhanced the stability of the polymer chains, resulting in requiring more energy to rupture the material. Furthermore, the PE-PP copolymer formed in the polymer blends modified with BPO and DVB, improved the compatibility of PE and PP.

Figure 3 shows the tensile strength and elongation at break of the polymer blends with different BPO content, and DVB content fixed at 0.5wt%; the corresponding values for the HDPE were 20.46MPa and 150%. The tensile strength increased slightly with increasing BPO content. With the addition of BPO and DVB, crosslinks were formed in the blends, which

enhanced the intermolecular forces. On the other hand, the formation of PE-PP copolymer, as above, improved the compatibility between PE and PP. Both of these increased the tensile strength of the polymer blends.

The elongation at break of the polymer blends increased with the BPO content up to 0.1wt%, but thereafter, its value decreased with the increase of BPO content. As the compatibility of the PE and PP was improved by the formation of copolymer, the elongation at break of the polymer blends increased with the addition of BPO and DVB. On the other hand, the increase of the chemical crosslinked structure intensified the restriction imposed on the elongational behavior of the polymer blends. This restriction is due to smaller segments lengths available for stretching and less probability of chain slippage, and resulted in the decrease in elongation at break above 0.1wt% BPO [16]. The elongation at break of the polymer blends showed the combined result of the two opposite effects.

Thermogravimetric analysis

Figure 4 shows the TGA curves of the unmodified polymer blends and the polymer blends modified with 0.4wt% BPO with and without DVB. The polymer blends did not show a major weight loss up to 400°C, after which the mass loss due to thermal decomposition began. The onset temperature of thermal decomposition decreased with the addition of BPO or BPO together with DVB. Crosslinking introduces tertiary carbons in the system, which are more prone to thermal decomposition [16]. Thus, the addition of BPO or BPO with DVB increased the number of the

tertiary carbon atoms resulting in reduced onset temperature of decomposition. Similar results were reported by Krupa and Luyt [17] and Khonakdar et al [16].

The differential thermogravimetry (DTG) curves are presented in Fig.5. There was one main peak in each curve whose peak value corresponded to the temperature where the decomposition velocity was the maximum for each material. The modified systems showed a decreased temperature of maximum decomposition rate compared with the unmodified polymer blends.

From the results of TG and DTG, the conclusion can be drawn that, different from irradiation and silane crosslinking [18-19], peroxide crosslinking of polyolefins did not increase their thermal stability.

SEM observations

The impact fracture surface morphology of the unmodified and modified polymer blends was observed by SEM. In the polymer blends without modification (Figure 6 (a)), two incompatible phases with clear phase segregated domains can be seen, indicating the poor interfacial adhesion. The PP phase is, presumably, the small spheres, with the HDPE phase being continuous. With the addition of 0.4wt% BPO (Figure 6 (b)), the compatibility of phases was improved by the formation of copolymer. This can cause the increase of impact strength of the polymer blends with the addition of BPO. Figure 6 (c) shows that more fiber structures were formed in the polymer blends modified with BPO and DVB, resulting in enhanced intermolecular forces. It is noted the fracture surfaces of all the three samples suggested that they were foams. This was further confirmed by the

density measurement, with density of 0.90g/cm^3 for the molded samples, which was lower than that of HDPE. The formation of foams would contribute to the low mechanical properties, although the cause of the forms is still under investigation.

Conclusions

Polymer blends of recycled HDPE and PP were modified with the peroxide BPO, which lowered their MFR and increased the impact strength. The addition of DVB further improved the degree of crosslinking as well as the impact strength. With fixed DVB content, an increase of BPO content enhanced the tensile strength and impact strength and decreased the melt flow rate of the polymer blends. The elongation at break increased with BPO content up to 0.1% and then decreased with further increase of BPO. The addition of BPO with or without DVB slightly reduced the thermostability of the polymer blends. The results of SEM showed that the addition of BPO with or without DVB improved the compatibility of PE and PP, resulting in the increased mechanical properties.

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Table 1 The melt flow rate and impact strength of PE/PP blends with and without BPO

Sample	MFR(g/10min)	Impact strength(kJ/m ²)
PE/PP (90/10)	0.350	7.3
PE/PP/BPO (90/10/0.4)	0.040	14.5

Table 2 The effect of DVB on the melt flow rate and impact strength of pure HDPE and BPO

modified PE/PP blends

Sample	MFR(g/10min)	Impact strength (kJ/m ²)
HDPE	0.33	19.0
PE/PP/BPO(90/10/0.4)	0.040	14.5
PE/PP/BPO/DVB(90/10/0.4/0.5)	0.025	21.0

Fig. 1. Effect of BPO content on the melt flow rate of PE/PP (90/10) blends; the DVB content was fixed at 0.5wt%.

Fig 2. Effect of BPO content on the impact strength of PE/PP (90/10) blends; the DVB content was fixed at 0.5wt%.

Fig. 3. Effect of BPO content on the tensile strength and elongation at break of PE/PP (90/10) blends; the DVB content was fixed at 0.5wt%.

Fig 4. Thermogravimetric curves of unmodified polyblends and BPO modified polyblends with and without DVB.

Fig. 5. DTG curves of unmodified polyblends and BPO modified polyblends with and without DVB.

Fig 6. SEM of impact fracture surfaces of PE/PP (90/10) blends (a) without modification; (b) modified with 0.4wt% BPO; and (c) modified with > 0.4wt% BPO together with 0.5wt% DVB.

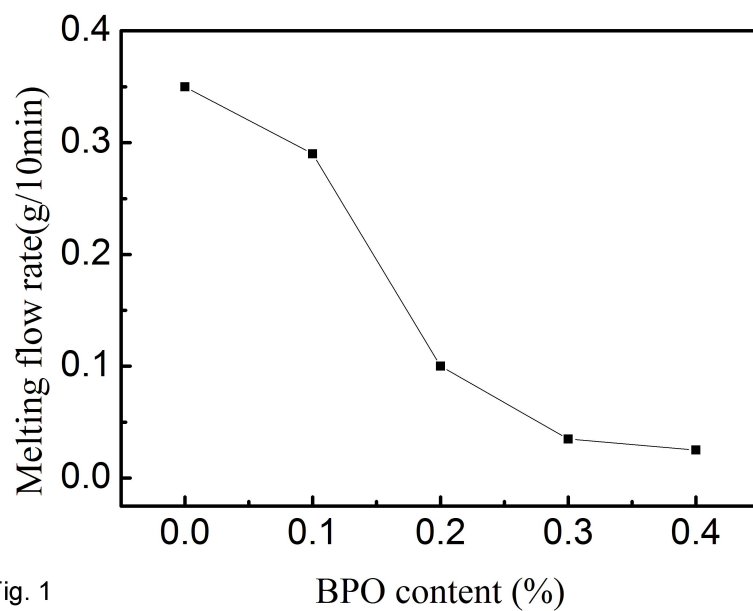
Figure 1.

Fig. 1

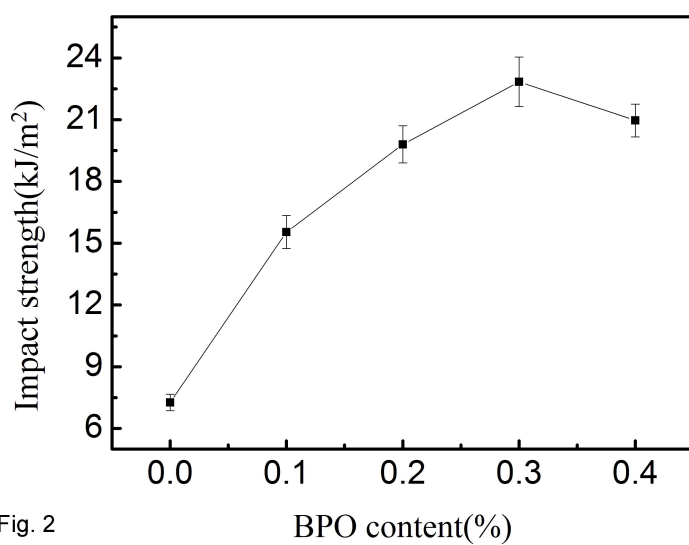
Figure 2.

Fig. 2

Figure 3.

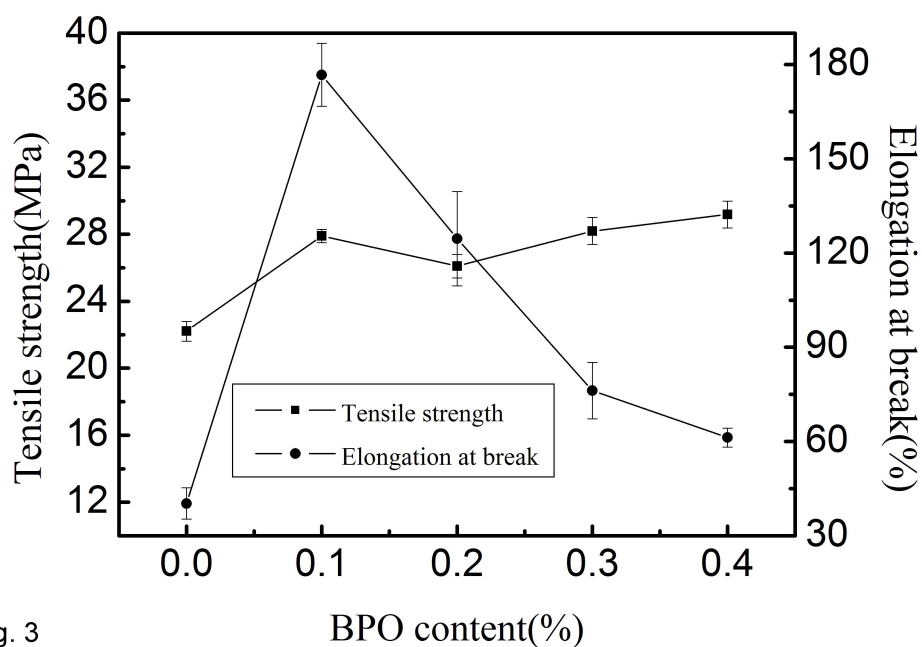


Fig. 3

Figure 4.

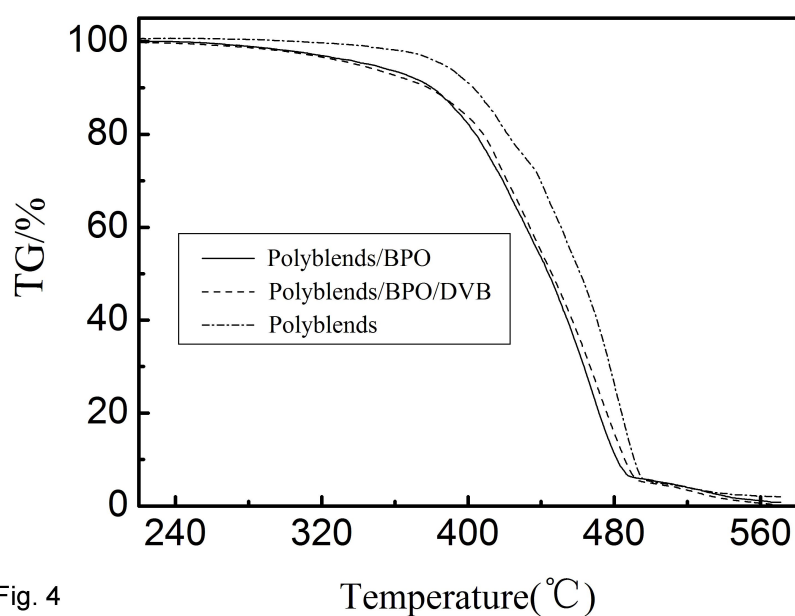


Fig. 4

Figure 5.

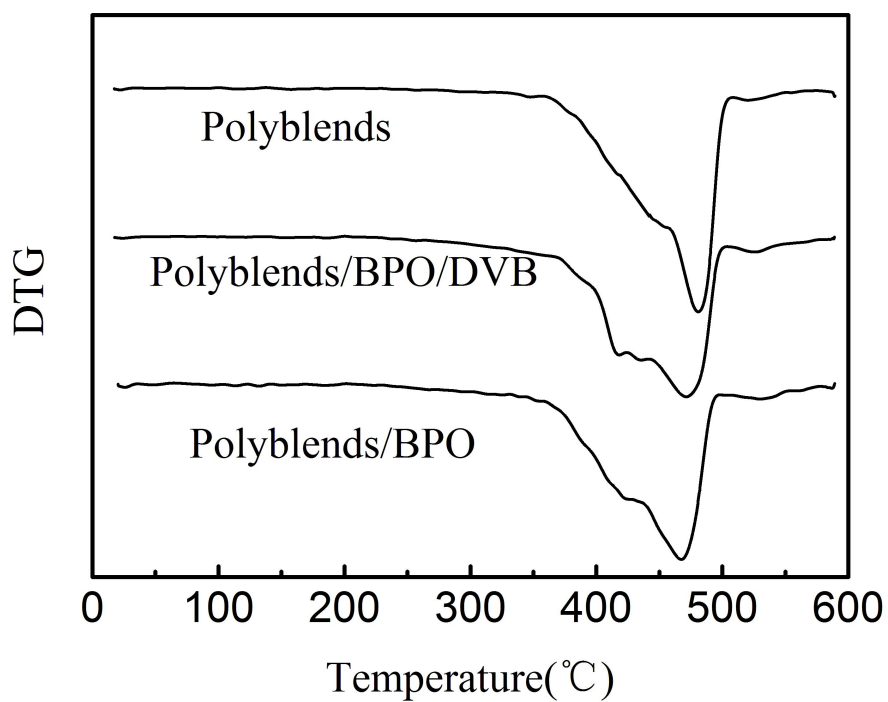


Fig. 5

Figure 6.

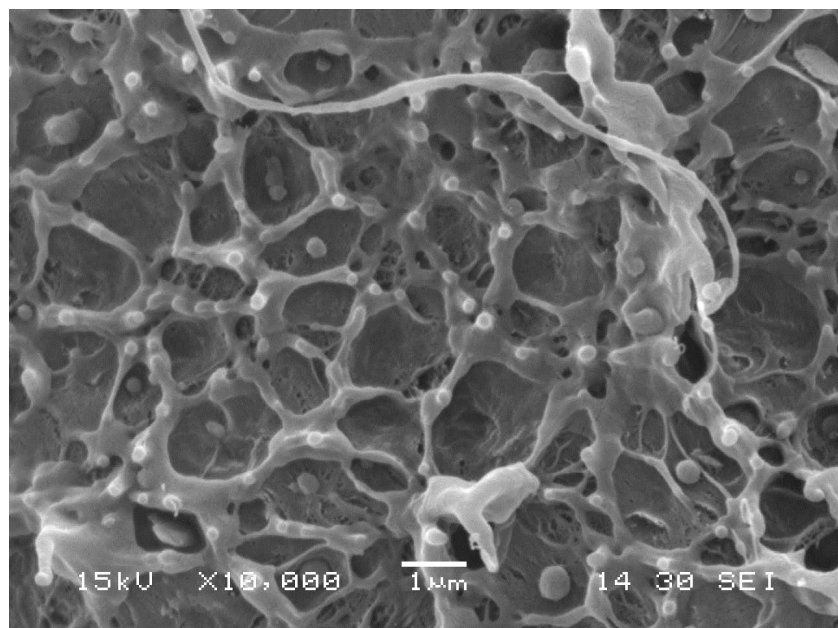


Fig. 6a

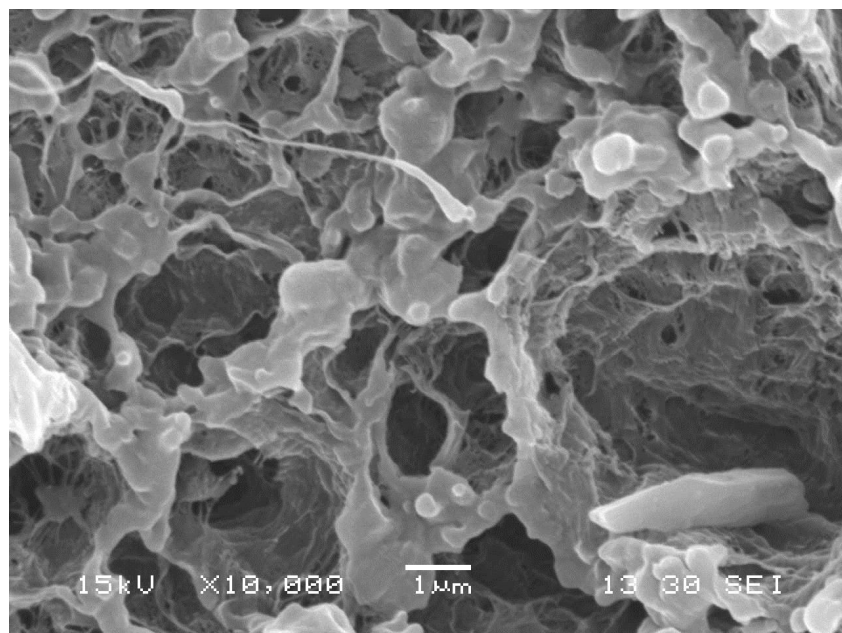


Fig. 6b

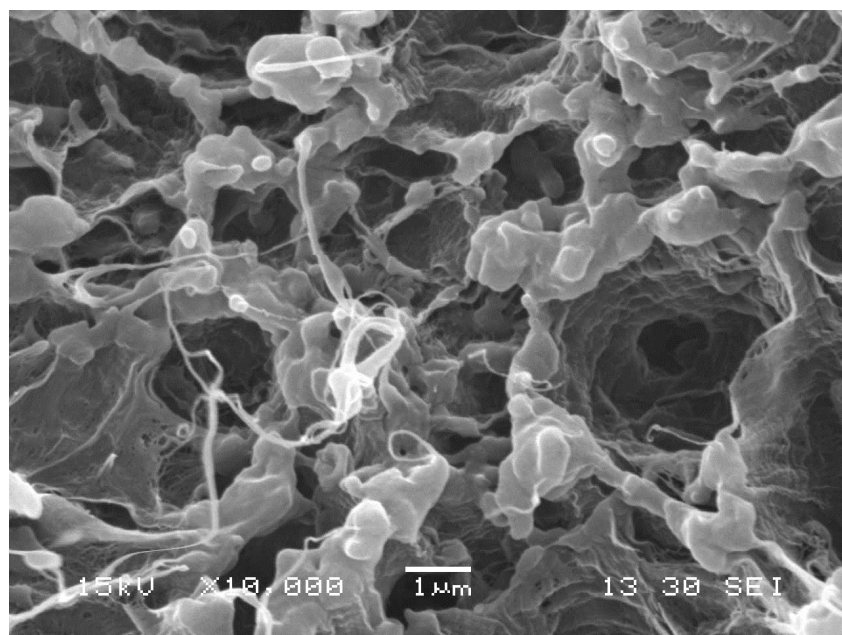


Fig. 6c